Practical: Creating GUI Forms Using Tkinter

Q. Create a Python GUI application using Tkinter to design a student registration form. The form should demonstrate the use of

labels, text boxes, buttons, menus, and scrollbars

. Handle widget properties and arrange the widgets using

layout management techniques

such as

`pack()`

,

`grid()`

, and

`place()`

.

File Name:

`student_registration.py`

Code:

```
import tkinter as tk
from tkinter import messagebox


# Create main window
root = tk.Tk()
root.title("Student Registration Form")
root.geometry("500x500")
root.resizable(False, False)


# Function to submit form
def submit_form():
    name = name_entry.get()
    email = email_entry.get()
    course = course_entry.get()

    if name == "" or email == "" or course == "":
        messagebox.showwarning("Warning", "Please fill all the fields.")
```

```
    else:
        messagebox.showinfo(
            "Registration Successful",
            "Name: " + name +
            "\nEmail: " + email +
            "\nCourse: " + course
        )


# Function to clear form
def clear_form():
    name_entry.delete(0, tk.END)
    email_entry.delete(0, tk.END)
    course_entry.delete(0, tk.END)


# ---------------- MENU ----------------

menubar = tk.Menu(root)

file_menu = tk.Menu(menubar, tearoff=0)
file_menu.add_command(label="Clear", command=clear_form)
file_menu.add_separator()
file_menu.add_command(label="Exit", command=root.destroy)

menubar.add_cascade(label="File", menu=file_menu)

help_menu = tk.Menu(menubar, tearoff=0)
help_menu.add_command(
    label="About",
    command=lambda: messagebox.showinfo(
        "About",
        "Student Registration Form\nCreated using Python Tkinter"
    )
)

menubar.add_cascade(label="Help", menu=help_menu)

root.config(menu=menubar)


# ---------------- TITLE ----------------

title_label = tk.Label(
    root,
    text="Student Registration Form",
    font=("Arial", 18, "bold")
)

title_label.pack(pady=15)


# ---------------- FORM FRAME ----------------

form_frame = tk.Frame(root)
form_frame.pack(pady=10)


# Name
name_label = tk.Label(
    form_frame,
```

```
    text="Name:",
    font=("Arial", 12)
)
name_label.grid(row=0, column=0, padx=10, pady=10, sticky="w")

name_entry = tk.Entry(
    form_frame,
    width=30
)
name_entry.grid(row=0, column=1, padx=10, pady=10)


# Email
email_label = tk.Label(
    form_frame,
    text="Email:",
    font=("Arial", 12)
)
email_label.grid(row=1, column=0, padx=10, pady=10, sticky="w")

email_entry = tk.Entry(
    form_frame,
    width=30
)
email_entry.grid(row=1, column=1, padx=10, pady=10)


# Course
course_label = tk.Label(
    form_frame,
    text="Course:",
    font=("Arial", 12)
)
course_label.grid(row=2, column=0, padx=10, pady=10, sticky="w")

course_entry = tk.Entry(
    form_frame,
    width=30
)
course_entry.grid(row=2, column=1, padx=10, pady=10)


# ---------------- BUTTONS ----------------

button_frame = tk.Frame(root)
button_frame.pack(pady=15)

submit_button = tk.Button(
    button_frame,
    text="Submit",
    width=12,
    command=submit_form
)

submit_button.grid(row=0, column=0, padx=10)

clear_button = tk.Button(
    button_frame,
    text="Clear",
    width=12,
    command=clear_form
```

```
)

clear_button.grid(row=0, column=1, padx=10)


# ---------------- SCROLLBAR ----------------

text_frame = tk.Frame(root)
text_frame.pack(pady=10)

scrollbar = tk.Scrollbar(text_frame)
scrollbar.pack(side=tk.RIGHT, fill=tk.Y)

text_box = tk.Text(
    text_frame,
    width=50,
    height=5,
    yscrollcommand=scrollbar.set
)

text_box.pack(side=tk.LEFT)

scrollbar.config(command=text_box.yview)


# Start GUI application
root.mainloop()
```

Output:

The application displays a **Student Registration Form** containing:

- **Label** – displays field names such as Name, Email and Course.
- **Text boxes (`Entry`)** – accept user input.
- **Buttons** – Submit and Clear.
- **Menu** – File and Help menus.
- **Text box (`Text`)** – allows multi-line text input.
- **Scrollbar** – scrolls through the contents of the text box.
- **Layout management** – `pack()` and `grid()` are used to arrange widgets.

Important Tkinter Widgets Used

Widget	Purpose
`Label`	Displays text
`Entry`	Accepts single-line input
`Button`	Performs an action
`Text`	Accepts multi-line input
`Menu`	Creates application menus
`Scrollbar`	Provides scrolling

`Frame`	Groups related widgets

Layout Management Used

`pack()`

Arranges widgets one after another.

`grid()`

Arranges widgets in rows and columns.

`place()`

Can position widgets at specific `x` and `y` coordinates.

Note: Avoid using `pack()` and `grid()` together **inside the same parent/container**. They can be used in different frames, as demonstrated above.